

Day 4: The SCIP LP Interface

Computational Integer Optimization · SoSe 2026

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Section 1

Simplex review

A numerical example

Consider the linear program

$$\begin{array}{ll}\max & 5x_1 + 4x_2 + 3x_3 \\ \text{s.t.} & 2x_1 + 3x_2 + x_3 \leq 5, \\ & 4x_1 + x_2 + 2x_3 \leq 11, \\ & 3x_1 + 4x_2 + 2x_3 \leq 8, \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Example adapted from Vanderbei, *Linear Programming*, Chapter 2, Section 1.

Add slack variables

A slack variable records the unused capacity in each constraint.

$$\begin{aligned}\zeta &= 5x_1 + 4x_2 + 3x_3, \\ w_1 &= 5 - 2x_1 - 3x_2 - x_3, \\ w_2 &= 11 - 4x_1 - x_2 - 2x_3, \\ w_3 &= 8 - 3x_1 - 4x_2 - 2x_3,\end{aligned}\quad x, w \geq 0.$$

- Basic variables: w_1, w_2, w_3 .
- Nonbasic variables: x_1, x_2, x_3 .
- Setting nonbasic variables to zero gives a basic solution.

Start from a basic feasible solution

Set the nonbasic variables to zero:

$$(x_1, x_2, x_3) = (0, 0, 0).$$

The dictionary then gives

$$(w_1, w_2, w_3) = (5, 11, 8), \quad \zeta = 0.$$



First iteration: choose an entering variable

The objective row is

$$\zeta = 5x_1 + 4x_2 + 3x_3.$$

All three variables could improve the objective. Using the largest positive objective coefficient, choose

x_1 enters the basis.

As x_1 increases from zero,

$$w_1 = 5 - 2x_1, \quad w_2 = 11 - 4x_1, \quad w_3 = 8 - 3x_1.$$

The basic variables must remain nonnegative.

First iteration: the ratio test

How far can x_1 increase?

$$w_1 \geq 0 \implies x_1 \leq 5/2,$$

$$w_2 \geq 0 \implies x_1 \leq 11/4,$$

$$w_3 \geq 0 \implies x_1 \leq 8/3.$$

Therefore,

$$x_1 = \min \left\{ \frac{5}{2}, \frac{11}{4}, \frac{8}{3} \right\} = \frac{5}{2}.$$

w_1 reaches zero first: w_1 leaves the basis.

Pivot: exchange x_1 and w_1

Solve the w_1 row for x_1 , then eliminate x_1 from the other rows:

$$\zeta = 12.5 - 2.5w_1 - 3.5x_2 + 0.5x_3,$$

$$x_1 = 2.5 - 0.5w_1 - 1.5x_2 - 0.5x_3,$$

$$w_2 = 1 + 2w_1 + 5x_2,$$

$$w_3 = 0.5 + 1.5w_1 + 0.5x_2 - 0.5x_3.$$

The new basic feasible solution is

$$(x_1, x_2, x_3) = \left(\frac{5}{2}, 0, 0\right), \quad (w_1, w_2, w_3) = \left(0, 1, \frac{1}{2}\right), \quad \zeta = 12.5.$$

Second iteration

In the new objective row,

$$\zeta = 12.5 - 2.5w_1 - 3.5x_2 + 0.5x_3,$$

only x_3 has a positive coefficient. Thus, x_3 enters.

Keeping $w_1 = x_2 = 0$ while increasing x_3 gives

$$x_1 = 2.5 - 0.5x_3, \quad w_2 = 1, \quad w_3 = 0.5 - 0.5x_3.$$

Hence

$$x_3 \leq 5, \quad x_3 \leq 1.$$

$x_3 = 1$ and w_3 leaves the basis.

Pivot again

Exchange x_3 and w_3 :

$$\zeta = 13 - w_1 - 3x_2 - w_3,$$

$$x_1 = 2 - 2w_1 - 2x_2 + w_3,$$

$$w_2 = 1 + 2w_1 + 5x_2,$$

$$x_3 = 1 + 3w_1 + x_2 - 2w_3.$$

Setting the nonbasic variables w_1, x_2, w_3 to zero gives

$$\boxed{x^* = (2, 0, 1)}, \quad (w_1, w_2, w_3) = (0, 1, 0), \quad \boxed{\zeta = 13}.$$

Why can we stop?

The final objective row is

$$\zeta = 13 - w_1 - 3x_2 - w_3.$$

Every nonbasic variable is constrained to be nonnegative. Therefore,

$$\zeta \leq 13$$

for every feasible solution represented by this equivalent dictionary.

Our current solution attains $\zeta = 13$, so it is optimal:

$$x^* = (2, 0, 1), \quad \zeta^* = 13.$$

Section 2

Obtaining LP information from SCIP

The LP seen by SCIP

SCIP internally represents the LP relaxation in the form

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax + Is = 0, \\ & -U \leq s \leq -L, \\ & \ell \leq x \leq u.\end{array}$$

A row $L_i \leq A_i x \leq U_i$ is represented using

$$s_i = -A_i x.$$

Thus, the row bounds become $-U_i \leq s_i \leq -L_i$.

Section 3

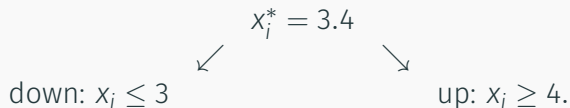
Driebeek penalties

Branching on a fractional basic variable

Suppose the LP relaxation gives an integer-constrained variable

$$x_i^* = 3.4.$$

Branch-and-bound creates two children:



The required movements are

$$d_i^- = 3.4 - 3 = 0.4, \quad d_i^+ = 4 - 3.4 = 0.6.$$

At a simplex basic feasible solution, a fractional integer variable is typically basic; the nonbasic variables sit at bounds.

An objective and its reduced costs

Consider the minimization objective

$$\min \quad z = x_1 + 5x_2 - 3x_3 + 2.5x_4.$$

For the current basis, SCIP's tableau row for basic variable x_1 is

$$0 = x_1 + 2x_2 + x_3 + 0.5x_4.$$

Eliminating x_1 from the objective gives the reduced-cost row:

$$\begin{aligned} z &= (-2x_2 - x_3 - 0.5x_4) + 5x_2 - 3x_3 + 2.5x_4 \\ &= \boxed{3x_2 - 4x_3 + 2x_4}. \end{aligned}$$

Thus, the reduced costs of (x_2, x_3, x_4) are $(3, -4, 2)$.

At the current LP solution

$$x^* = (3.4, 0, -3.4, 0), \quad z^* = 3.4 - 3(-3.4) = 13.6.$$

Then inspect the zero-right-hand-side tableau row

Suppose SCIP returns

$$0 = x_1 + 2x_2 + x_3 + 0.5x_4.$$

The current LP solution and reduced costs are:

Variable	LP value	Status	Reduced cost
x_1	3.4	BASIC	0
x_2	0	LOWER	3
x_3	-3.4	UPPER	-4
x_4	0	LOWER	2

Indeed, the current values satisfy the raw tableau row:

$$3.4 + 2(0) - 3.4 + 0.5(0) = 0.$$

We now test which single feasible movement reaches each branch bound.

Down branch: $x_1 \leq 3$

We must decrease x_1 by 0.4. The lower-bound variables x_2 and x_4 can move upward.

Move x_2 from its lower bound

$$2\delta_2 = 0.4 \implies \delta_2 = 0.2.$$

New solution and objective:

$$x' = (3, 0.2, -3.4, 0),$$

$$z' = 3 + 5(0.2) - 3(-3.4) = 14.2.$$

Move x_4 from its lower bound

$$0.5\delta_4 = 0.4 \implies \delta_4 = 0.8.$$

New solution and objective:

$$x' = (3, 0, -3.4, 0.8),$$

$$z' = 3 - 3(-3.4) + 2.5(0.8) = 15.2.$$

$$p_1^{\text{down}} = \min\{14.2 - 13.6, 15.2 - 13.6\} = 0.6.$$

Up branch: $x_1 \geq 4$

We must increase x_1 by 0.6. Variable x_3 is at its upper bound -3.4 , so it may move downward:

$$-\delta_3 = 0.6 \implies \delta_3 = -0.6, \quad x_3 = -4.0.$$

The new solution and its objective value are

$$x' = (4, 0, -4, 0),$$

$$z' = 4 - 3(-4) = 16.$$

Therefore,

$$P_1^{\text{up}} = 16 - 13.6 = 2.4.$$

The signs are essential: a negative reduced cost times a feasible negative movement gives a nonnegative penalty.